

ORANGE COUNTY REGIONAL HISTORY CENTER HVAC REPLACEMENT



FINAL CONSTRUCTION DOCUMENTS

FEBRUARY 10, 2015

ATKINS

**482 SOUTH KELLER ROAD
ORLANDO, FLORIDA 32810**

Sheet Number	Sheet Name	Scale	Sheet Issues
M-000	COVER SHEET AND SHEET INDEX		Yes
M-001	HVAC SYMBOLS LEGEND AND GENERAL NOTES	No Scale	Yes
M-101	HVAC DEMOLITION PLAN ROOF PLAN	1/8"=1'-0"	Yes
M-102	HVAC NEW WORK PLAN ROOF PLAN	1/8"=1'-0"	Yes
M-103	HVAC PHASING PLAN	1/8"=1'-0"	Yes
M-201	HVAC CONTROLS	No Scale	Yes
M-301	HVAC DETAILS	No Scale	Yes
M-401	HVAC SCHEDULES	No Scale	Yes
S-500	STRUCTURAL SUPPORT DETAILS	No Scale	Yes

ATKINS

**482 SOUTH KELLER ROAD
ORLANDO, FL 32810
PHONE: 407.647.7275 FAX: 407.740.8958**

CONSULTANT:

CLIENT:

PROJECT NAME:

Orange County Regional History Center

65 E Central Blvd, Orlando, FL 32801

100040229

[illegible]

ISSUE LOG

PROFESSIONAL SEALS:

Thomas J. Farmer
FL PE # 58890

SHEET TITLE:

**COVER SHEET AND SHEET
INDEX**

SHEET INFORMATION:

JOB No. 100040229	Date Issued: 2/10/15
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Modeled By: REP

Sheet Number:	
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Checked By:	DLH
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1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107	108	109	110	111	112	113	114	115	116	117	118	119	120	121	122	123	124	125	126	127	128	129	130	131	132	133	134	135	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165	166	167	168	169	170	171	172	173	174	175	176	177	178	179	180	181	182	183	184	185	186	187	188	189	190	191	192	193	194	195	196	197	198	199	200	201	202	203	204	205	206	207	208	209	210	211	212	213	214	215	216	217	218	219	220	221	222	223	224	225	226	227	228	229	230	231	232	233	234	235	236	237	238	239	240	241	242	243	244	245	246	247	248	249	250	251	252	253	254	255	256	257	258	259	260	261	262	263	264	265	266	267	268	269	270	271	272	273	274	275	276	277	278	279	280	281	282	283	284	285	286	287	288	289	290	291	292	293	294	295	296	297	298	299	300	301	302	303	304	305	306	307	308	309	310	311	312	313	314	315	316	317	318	319	320	321	322	323	324	325	326	327	328	329	330	331	332	333	334	335	336	337	338	339	340	341	342	343	344	345	346	347	348	349	350	351	352	353	354	355	356	357	358	359	360	361	362	363	364	365	366	367	368	369	370	371	372	373	374	375	376	377	378	379	380	381	382	383	384	385	386	387	388	389	390	391	392	393	394	395	396	397	398	399	400	401	402	403	404	405	406	407	408	409	410	411	412	413	414	415	416	417	418	419	420	421	422	423	424	425	426	427	428	429	430	431	432	433	434	435	436	437	438	439	440	441	442	443	444	445	446	447	448	449	450	451	452	453	454	455	456	457	458	459	460	461	462	463	464	465	466
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QC Review: T.JF

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M-000

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65 E Central Blvd, Orlando, FL 32801

100040229

[illegible]

ISSUE LOG

PROFESSIONAL SEALS

Thomas J. Farmer
FL PE # 58890

SHEET TITLE

HVAC SYMBOLS LEGEND AND GENERAL NOTES

SHEET INFORMATION:

JOB No. 100040229	Date Issued: 2/10/1
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Modeled By:	RFP
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Checked By	Date

Checked by: DLH

QC Review: TJF

M-001

HVAC SYMBOL LEGEND				HVAC ABBREVIATIONS				HVAC GENERAL NOTES	
SYMBOL	DESCRIPTION	SYMBOL	DESCRIPTION	SYMBOL	DESCRIPTION	SYMBOL	DESCRIPTION	SYMBOL	DESCRIPTION
	-REVISION REFERENCE		-OUTSIDE AIR LOUVER		-1" INTERNALLY LINED DUCTWORK	AFD	-ADJUSTABLE FREQUENCY DRIVE	MAU	-MAKE-UP AIR UNIT
	-DETAIL REFERENCE: TOP DETAIL, BOTTOM-DRAWING WORK ON		-EXHAUST AIR LOUVER		-HYDRONIC REHEAT COIL	AFR	-ABOVE FINISHED FLOOR	MBH	-THOUSAND BTU PER HOUR
	-THERMOSTAT/TEMPERATURE SENSOR		-DUCTWORK SOUND ATTENUATOR		-INLINE CENTRIFUGAL FAN	AFR	-ABOVE FINISHED ROOF	MCA	-MINIMUM CIRCUIT AMPS
	-HUMIDISTAT/HUMIDITY SENSOR		-DOOR GRILLE		-CHANGE OF ELEVATION	AHU	-AIR HANDLING UNIT	MOC	-MAXIMUM OVER CURRENT PROTECTION
	-EMERGENCY SWITCH		-UNDERCUT DOOR		-FLEXIBLE DUCT	AP	-ACCESS PANEL	MOD	-MODULATOR OPERATED CONTROL DAMPER (MOD)
	-CO2 SENSOR		-ACCESS DOORS, VERTICAL OR HORIZONTAL		-TRANSITION, CONCENTRIC	BOP	-BOTTOM OF PIPE	MTHW	-MEDIUM TEMP HOT WATER
	-DUCT SMOKE DETECTOR		-FLAT OVAL DUCT		-TRANSITION, ECCENTRIC	BHP	-BRAKE HORSEPOWER	NC	-NORMALLY CLOSED
	-CONNECT TO EXISTING		-NEW DUCTWORK, FIRST DIMENSION IS SIDE SHOWN		-TRANSITION, SQUARE TO ROUND	BHPW	-BUILDING HOT WATER PUMP	NO	-NORMALLY OPEN
	-DEMOLISH TO POINT INDICATED		-DUCT ELBOW, POSITIVE PRESSURE (SUPPLY), FIRST		-SQUARE THROAT ELBOW W/TURNING VANES	BTU	-BRITISH THERMAL UNIT	NTS	-NOT TO SCALE
	-MOTORIZED CONTROL DAMPER		-DUCT ELBOW, EXHAUST		-RADIUS ELBOW	C	-CENTER LINE	OA	-OUTSIDE AIR
	-TEMPERATURE SENSOR		-DUCT ELBOW, NEGATIVE PRESSURE, RETURN		-RECTANGULAR/ROUND CONICAL TAKE-OFF OR ROUND/ROUND CONICAL TAKE-OFF	CFM	-CFM (CUBIC FEET PER MINUTE)	OAL	-OUTSIDE AIR LOUVER
	-PRESSURE SENSOR		-DUCT ELBOW, UP THROUGH ROOF OR SLAB ABOVE		-SQUARE THROAT TEE	CD	-CEILING DIFFUSER	PRBP	-PRESSURE REDUCING BACKFLOW PREVENTER
	-BACKDRAFT DAMPER		-DUCT ELBOW, UP THROUGH SLAB W/FAN ON ROOF ABOVE		-RADIUS TEE	CT	-CONSTANT AIR VOLUME	PRV	-PRESSURE REDUCING VALVE
	-SHEET NOTE CALLOUT		-DUCT ELBOW, NEGATIVE PRESSURE, RETURN		-RECTANGULAR DUCT SECTION UP, POSITIVE PRESSURE, SUPPLY OR OUTSIDE AIR	CTP	-CHANGE IN PRESSURE	PRS	-PRESSURE REDUCING STATION
	-SHEET NOTE CALLOUT		-DUCT ELBOW, UP THROUGH ROOF OR SLAB ABOVE		-RECTANGULAR DUCT SECTION UP, NEGATIVE PRESSURE, RETURN	CTP	-CHANGE IN PRESSURE	PSI	-POUNDS PER SQUARE INCH
	-AIR DISTRIBUTION TAG		-DUCT ELBOW, UP THROUGH ROOF OR SLAB ABOVE		-RECTANGULAR DUCT SECTION UP, EXHAUST	CTP	-CHANGE IN PRESSURE	PSIG	-PSI GAUGE
	-CEILING MOUNTED ACCESS DOOR		-DUCT ELBOW, UP THROUGH ROOF OR SLAB ABOVE		-ROUND DUCT SECTION UP	CTP	-CHANGE IN PRESSURE	PSI	-PACKAGED TERMINAL AIR CONDITIONER
	-CEILING DIFFUSER, ROUND OR RECTANGULAR NECK (CEILING DIFFUSERS ARE 4-WAY THROW UNO)		-DUCT ELBOW, UP THROUGH ROOF OR SLAB ABOVE		-FLAT OVAL DUCT SECTION UP	CU	-CONDENSING UNIT	PVC	-POLYVINYL CHLORIDE PIPE
	-ROUND DIFFUSER		-DUCT ELBOW, UP THROUGH ROOF OR SLAB ABOVE		-EXHAUST DUCT UP THROUGH SLAB W/FAN ON ROOF ABOVE	DD	-DIRECT DIGITAL CONTROLS	RA	-RETURN AIR
	-CEILING RETURN		-DUCT ELBOW, UP THROUGH ROOF OR SLAB ABOVE		-EXHAUST DUCT UP THROUGH ROOF W/DUCT DOWN THROUGH ROOF	DN	-DOWN	RHC	-REHEAT COIL
	-CEILING EXHAUST		-DUCT ELBOW, UP THROUGH ROOF OR SLAB ABOVE		-EXHAUST DUCT UP THROUGH ROOF W/DUCT DOWN THROUGH ROOF	DS	-DUCT SOUND ATTENUATOR	RHP	-ROOFTOP HEAT PUMP
	-CEILING DIFFUSER, RECTANGULAR (CEILING DIFFUSERS ARE 4-WAY THROW UNO)		-DUCT ELBOW, UP THROUGH ROOF OR SLAB ABOVE		-EXHAUST DUCT UP THROUGH ROOF W/DUCT DOWN THROUGH ROOF	EAT	-ENTERING AIR TEMPERATURE	RML	-REVOLUTIONS PER MINUTE
	-SUPPLY REGISTER OR GRILLE (VERTICAL, MANUAL, SIDEWALL)		-DUCT ELBOW, UP THROUGH ROOF OR SLAB ABOVE		-EXHAUST DUCT UP THROUGH ROOF W/DUCT DOWN THROUGH ROOF	EDH	-ELECTRIC DUCT HEATER	RS	-REFRIGERANT SUCTION & LIQUID LINES
	-RETURN EXHAUST REGISTER OR GRILLE (VERTICAL, MANUAL, SIDEWALL)		-DUCT ELBOW, UP THROUGH ROOF OR SLAB ABOVE		-EXHAUST DUCT UP THROUGH ROOF W/DUCT DOWN THROUGH ROOF	EF	-EXHAUST FAN	RTU	-ROOFTOP AIR HANDLING UNIT
	-SLEEVE (AS REQUIRED BY AR-19-1)		-DUCT ELBOW, UP THROUGH ROOF OR SLAB ABOVE		-EXHAUST DUCT UP THROUGH ROOF W/DUCT DOWN THROUGH ROOF	ERV	-ENERGY RECOVERY VENTILATOR	SA	-SUPPLY AIR
	-FIRE DAMPER (WITH ACCESS PANEL)		-DUCT ELBOW, UP THROUGH ROOF OR SLAB ABOVE		-EXHAUST DUCT UP THROUGH ROOF W/DUCT DOWN THROUGH ROOF	ESP	-EXTERNAL STATIC PRESSURE	SF	-SUPPLY FAN
	-FIRE & SMOKE DAMPER (WITH ACCESS PANEL)		-DUCT ELBOW, UP THROUGH ROOF OR SLAB ABOVE		-EXHAUST DUCT UP THROUGH ROOF W/DUCT DOWN THROUGH ROOF	SS	-STATIC PRESSURE	SS	-STAINLESS STEEL
	-MANUAL BALANCING DAMPER		-DUCT ELBOW, UP THROUGH ROOF OR SLAB ABOVE		-EXHAUST DUCT UP THROUGH ROOF W/DUCT DOWN THROUGH ROOF	FCU	-FAN COIL UNIT	TAB	-TEST AND BALANCE
	-DEMOLISHED/REMOVED DUCTWORK, PIPING AND/OR EQUIPMENT		-DUCT ELBOW, UP THROUGH ROOF OR SLAB ABOVE		-EXHAUST DUCT UP THROUGH ROOF W/DUCT DOWN THROUGH ROOF	FD	-FIRE DAMPER	TAD	-TRANSFER AIR DUCT
	-PIPING/DUCTWORK SUPPORT		-DUCT ELBOW, UP THROUGH ROOF OR SLAB ABOVE		-EXHAUST DUCT UP THROUGH ROOF W/DUCT DOWN THROUGH ROOF	FF	-FINAL FILTERS	TS	-TOTAL STATIC PRESSURE
			-TERMINAL UNIT, VARIABLE/CONSTANT AIR VOLUME WITH HOT WATER HEAT		-SPIN-IN TAKE-OFF W/VOLUME DAMPER & FLEXIBLE DUCT	FLA	-FULL LOAD AMPS	UH	-UNIT HEATER
			-TERMINAL UNIT, VARIABLE/CONSTANT AIR VOLUME WITH HOT WATER HEAT		-SPIN-IN TAKE-OFF W/VOLUME DAMPER & ROUND DUCT	FFM	-FEET PER MINUTE	UN	-UNLESS NOTED OTHERWISE
			-TERMINAL UNIT, VARIABLE/CONSTANT AIR VOLUME WITH HOT WATER HEAT		-SPIN-IN TAKE-OFF W/VOLUME DAMPER & ROUND DUCT	GRM	-GALLONS PER MINUTE	UN	-UNLESS NOTED OTHERWISE
			-TERMINAL UNIT, VARIABLE/CONSTANT AIR VOLUME WITH HOT WATER HEAT		-SPIN-IN TAKE-OFF W/VOLUME DAMPER & ROUND DUCT	IPH	-INCHES PER HOUR	VPH	-VOLTS/PHASE
			-TERMINAL UNIT, VARIABLE/CONSTANT AIR VOLUME WITH HOT WATER HEAT		-SPIN-IN TAKE-OFF W/VOLUME DAMPER & ROUND DUCT	KW	-KILOWATT	VAV	-VARIABLE AIR VOLUME
			-TERMINAL UNIT, VARIABLE/CONSTANT AIR VOLUME WITH HOT WATER HEAT		-SPIN-IN TAKE-OFF W/VOLUME DAMPER & ROUND DUCT	LAV	-LEAVING AIR TEMPERATURE	VDF	-VARIABLE FREQUENCY DRIVE
			-TERMINAL UNIT, VARIABLE/CONSTANT AIR VOLUME WITH HOT WATER HEAT		-SPIN-IN TAKE-OFF W/VOLUME DAMPER & ROUND DUCT	LWT	-LEAVING WATER TEMPERATURE	VTU	-VARIABLE VOLUME TERMINAL UNIT
			-TERMINAL UNIT, VARIABLE/CONSTANT AIR VOLUME WITH HOT WATER HEAT		-SPIN-IN TAKE-OFF W/VOLUME DAMPER & ROUND DUCT	LD	-LINEAR DIFFUSER		

HVAC PIPING SYMBOL LEGEND			
SYMBOL	DESCRIPTION	SYMBOL	DESCRIPTION
	-CHILLED WATER SUPPLY		-CONTROL VALVE
	-CHILLED WATER RETURN		-CHECK VALVE
	-CONDENSATE		-CALIBRATING BALANCING VALVE
	-CONDENSATE RETURN		-GAS COCK
	-PUMPED CONDENSATE		-UNION
	-HOT WATER SUPPLY		-STRAINER
	-HOT WATER RETURN		-PSI REG
	-BUILDING HOT WATER SUPPLY		-FLOW SWITCH
	-BUILDING HOT WATER RETURN		-SLOPE DIRECTION (DOWN)
	-MEDIUM TEMPERATURE WATER SUPPLY		-FLEX CONNECTION
	-MEDIUM TEMPERATURE WATER RETURN		-O.S.&Y. GATE VALVE
	-HIGH PRESSURE STEAM SUPPLY		-STEAM TRAP
	-HIGH PRESSURE STEAM RETURN		-THREE-WAY CONTROL VALVE
	-MEDIUM PRESSURE STEAM SUPPLY		-THERMOMETER
	-MEDIUM PRESSURE STEAM RETURN		-P-T RAP
	-LOW PRESSURE STEAM SUPPLY		-TWO-WAY CHECK VALVE
	-LOW PRESSURE STEAM RETURN		-MANUAL VENT
	-REFRIGERANT LIQUID		-PRESSURE GAUGE
	-REFRIGERANT SUCTION		-RELIEF VALVE
	-FLOW DIRECTION		-FLOW METER
	-GATE VALVE		-WATER METER
	-BALL VALVE		
	-BUTTERFLY VALVE		

SYMBOL	DESCRIPTION	SYMBOL	DESCRIPTION
	-BINARY / DIGITAL INPUT		-BINARY / DIGITAL OUTPUT
	-ANALOG INPUT		-ANALOG OUTPUT
	-DIFFERENTIAL PRESSURE SWITCH (DPS) OR TRANSMITTER (DPT)		-VARIABLE FREQUENCY DRIVE
	-FLOW METER (FM)		-PRESSURE / TEMPERATURE (PT) PORT
	-TEMPERATURE SENSOR IN WELL		-DUCT MOUNTED HUMIDITY SENSOR
	-DUCT MOUNTED TEMPERATURE SENSOR		

SYMBOL	DESCRIPTION	SYMBOL	DESCRIPTION
	-MOTORIZED 2-POSITION, FLOATING OR MODULATING CONTROL VALVE AS INDICATED		-PRESSURE SWITCH
	-MOTORIZED CONSTANT FLOW 3-WAY CONTROL VALVE		-CENTRIFUGAL FAN OR PUMP
	-DAMPER AND ACTUATOR WITH END SWITCH		-THERMOMETER GAUGE
	-STARTER DISCONNECT		-PRESSURE GAUGE
	-AIR FLOW MONITORING STATION ALONE OR AIR FLOW MONITORING STATION WITH 2-POSITION OR MODULATING MOTORIZED DAMPER		-CLOSED DIAPHRAGM TYPE EXPANSION TANK (ET)
	-CO2 SENSOR		-AIR SEPARATOR (AS)
	-WALL-MOUNTED THERMOSTAT / TEMP. SENSOR, HUMIDITY SENSOR OR CO2 SENSOR		-DRAIN VALVE WITH END PLUG
	-EVAPORATIVE COOLER		-MANUAL AIR VENT
	-HIGH / LOW STATIC PRESSURE SWITCH		-AUTOMATIC AIR VENT
			-AUTOMATIC FLOW CONTROL VALVE (AFCV) CIRCUIT SETTER
			-PRESSURE INDEPENDENT CONTROL VALVE (PICV)
			-FLEX CONNECTION

NOTE: SOME SYMBOLS SHOWN ON THIS LEGEND MAY NOT PERTAIN TO THIS PROJECT

CONSULTANT

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[illegible]

ISSUE LOG

PROFESSIONAL SEALS:

Thomas J. Farmer
FL PE # 58890

SHEET TITLE:

HVAC DEMOLITION PLAN ROOF PLAN

SHEET INFORMATION:

JOB No. 100040229	Date Issued: 2/10/1
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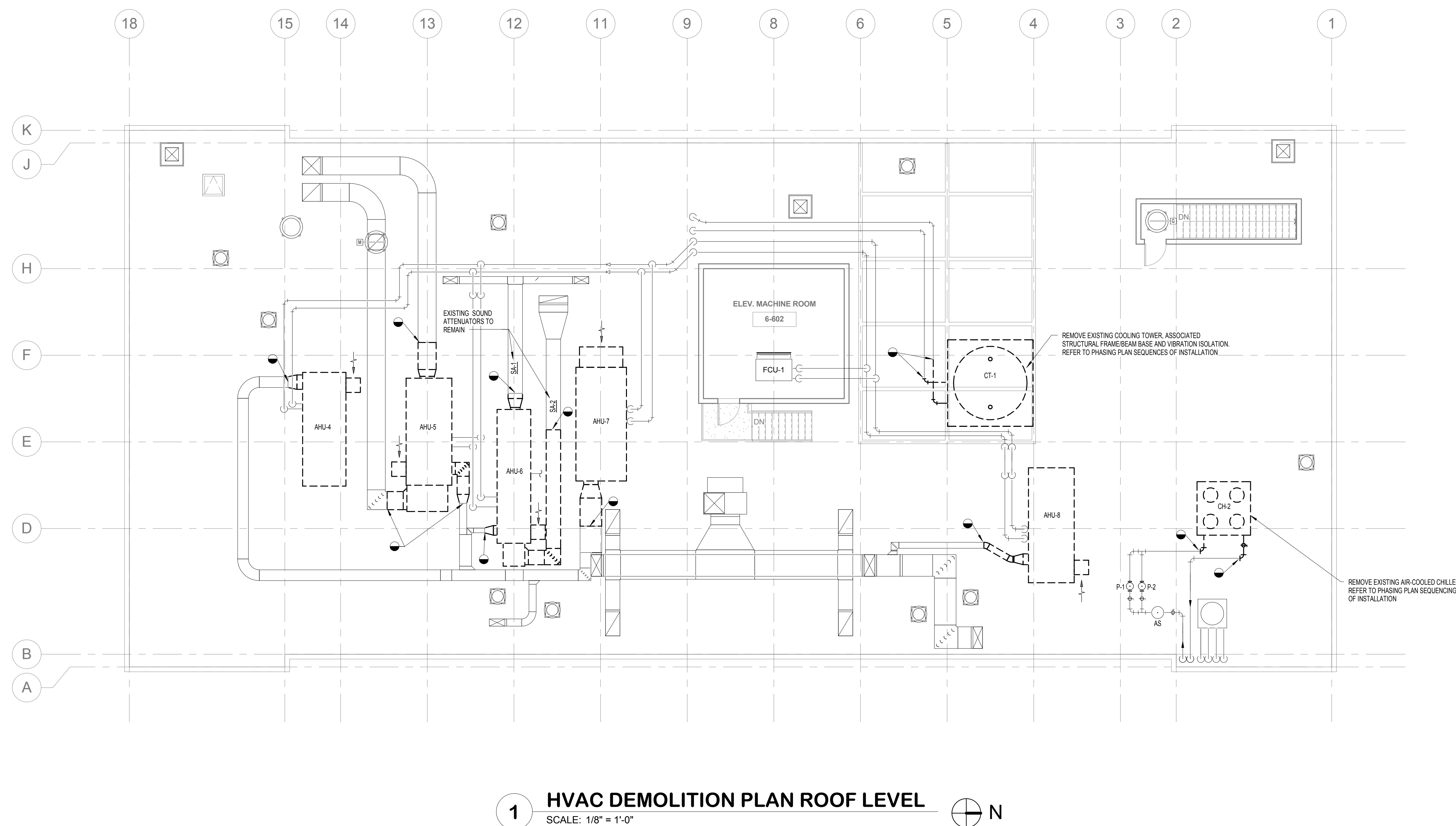
Modeled By: BFP

Modeled By:	REF
Checked By:	DLL

Checked By:	DLH
CC-BY:	DLH

QC Review:	TJF
By:	TJF

M-101



CONSULTANT:

CLIENT:

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PROFESSIONAL SEALS:

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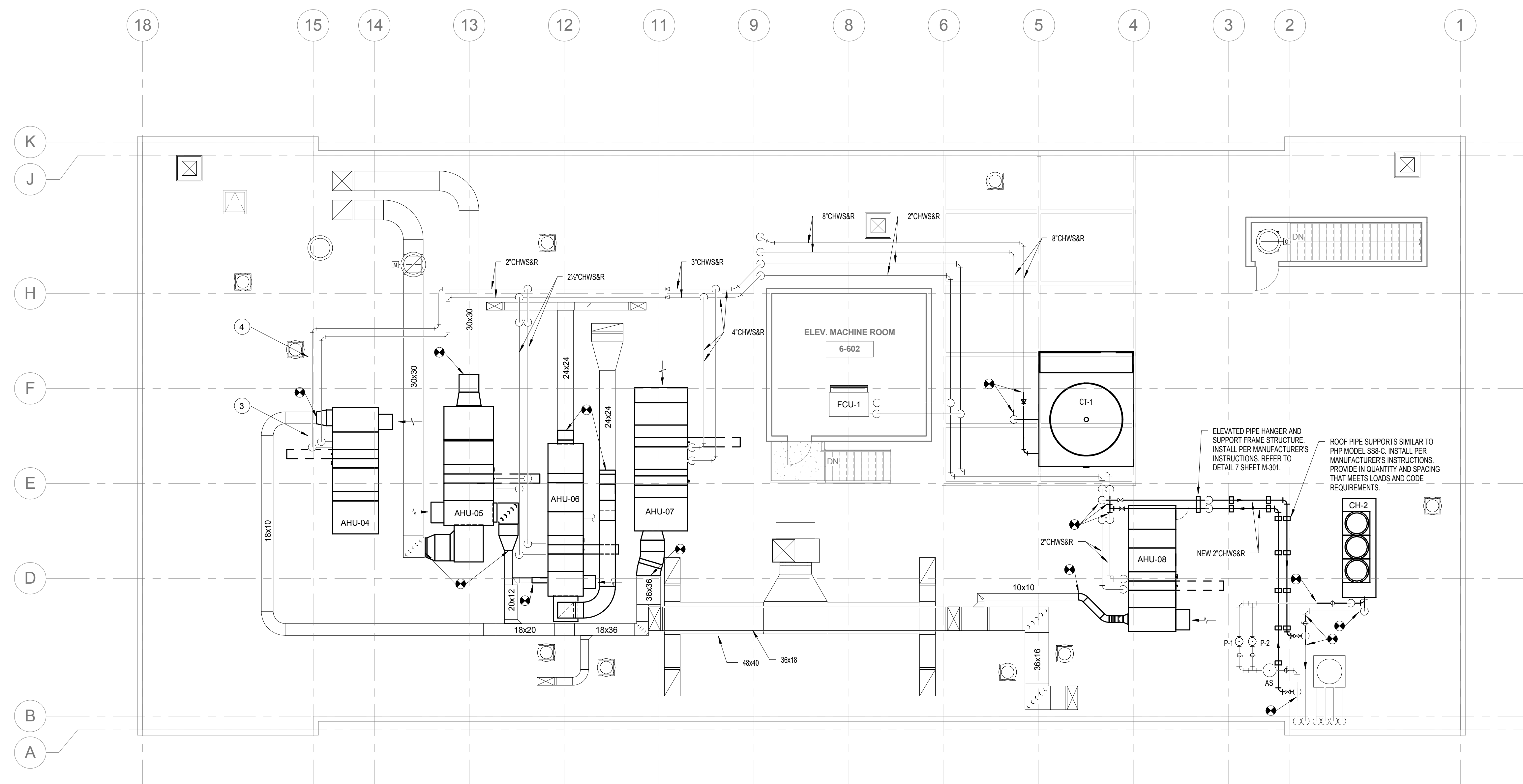
HVAC NEW WORK PLAN

ROOF PLAN

SHEET INFORMATION:		Date Issued: 2/10/11 Sheet Number: M-102
JOB NO. 100040229		
Modeled By: REP		
Checked By: DLH		
QC Review: TJF		
Phase: CD		

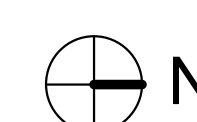
NEW WORK NOTES:

1. NEW EQUIPMENT SHOWN IN HEAVY LINE WEIGHT. EXISTING ROO CURBS AND EQUIP SUPPORTS SHALL BE RE-CONFIGURED TO SERVE AND WORK WITH NEW EQUIPMENT DIMENSIONS AND WEIGHTS. WORK SHALL INCLUDE ITEMS SUCH AS COUNT-ER-FLASHING COVERS, NEW WATER NAILER, NEW ANTI-MICROBIAL INSULATION, BLANKED-OFF SECTIONS, CURB TRANSITIONS, ETC.
2. EXISTING PIPES AND DUCTS SHALL BE RECONNECTED AS SHOWN.
3. REPAIR AND REPLACE ALL AFFECTED EQUIPMENT, DUCT, PIPING, ETC.
4. PROVIDE WATER / AIR-TIGHT CONNECTIONS TO NEW AND/OR EXISTING EQUIPMENT.
5. NEW PIPE/DUCT INSULATION AND JACKETING SHALL MATCH EXISTING (REFER TO SPECIFICATIONS)
6. EXISTING CONTROL WIRING SHALL BE RECONNECTED TO NEW AIR-HANDLING UNITS' FACTORY-INSTALLED INTERNAL DAMPER ACTUATORS(S) AND SENSORS.
7. EXISTING POWER WIRING SHALL BE RECONNECTED TO NEW AIR-HANDLING UNITS' FAN-MOTOR STARTER-DISCONNECT(S) AND/OR VFD(S).
8. EXISTING POWER WIRING SHALL BE RECONNECTED TO NEW ELECTRIC HUMIDIFIER.
9. PROVIDE EXISTING MAKE-UP-WATER CONNECTION TO NEW HUMIDIFIERS (PER MFG'S INSTRUCTIONS).
10. REMOVE NEW AHUS' CONDENSATE DRAIN P-TRAPS. RE-USE DRAIN LINES AS MUCH AS POSSIBLE.
11. REPLACE CHW COIL INLET STRAINER FILTER MESH.



1 HVAC NEW WORK PLAN ROOF LEVEL

SCALE: 1/8" = 1'-0"

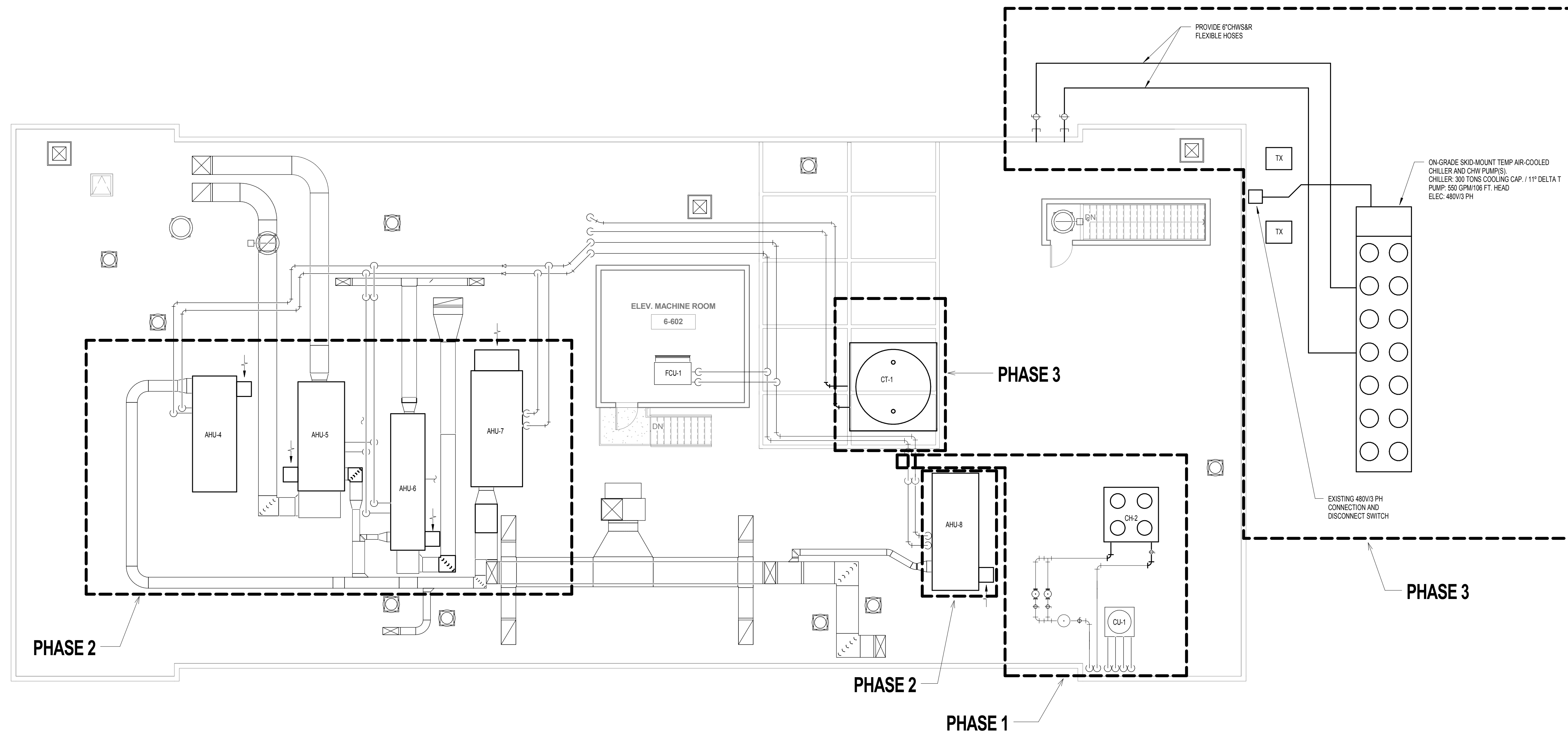


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SCALE: 1/8" = 1'-0"

PHASE 2:
PROVIDE FOR THE SEQUENTIAL REMOVAL AND REPLACEMENT OF AHUS 4, 5, 6, 7 & 8. CONTRACTOR SHALL CONSULT, COORDINATE AND SCHEDULE ORDER OF REPLACEMENT WITH BUILDING MAINTENANCE PERSONNEL AND/OR OCCUPANTS. (I.E. WORK MAY BE SCHEDULED TO TAKE ADVANTAGE OF LOW OUTDOOR AMBIENT TEMPERATURES AND/OR TAKE PLACE DURING UNOCCUPIED HOURS). CONTRACTOR SHALL DETERMINE IF THE USE OF INDOOR PORTABLE COOLING EQUIPMENT IS NECESSARY. THE FOLLOWING ORANGE COUNTY REGIONAL HISTORIC CENTER TEMPERATURE AND HUMIDITY GUIDELINES SHALL BE MAINTAINED THROUGHOUT CONSTRUCTION:

TEMPERATURE & HUMIDITY GUIDELINES					
EQUIP.	AREA/SPACE	TEMPERATURE		HUMIDITY	
		IDEAL	ACCEPT	IDEAL	ACCEPT
AHU-5	COURTROOM	70°	68°-72°	45%	40%-50%
AHU-8	THEATER	70°	60°-80°	45%	35%-55%
AHU-8	5TH FL. ARCHIVES	70°	68°-72°	35%	30%-45%
AHU-7	4TH FL. CHAP. GALLERY	70°	68°-72°	45%	40%-50%
CH-2	TRAVELING EXHIBIT HALL	70°	68°-72°	45%	40%-50%
AHU-7	ALL OTHER MUSEUM GALLERIES	70°	68°-75°	55%	45%-60%

[illegible]

PROFESSIONAL SEALS:

Thomas J. Farmer
FL PE # 58890

HVAC PHASING PLAN

JOB No. 100040229	Date Issued: 2/10/15
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Modeled By: REP

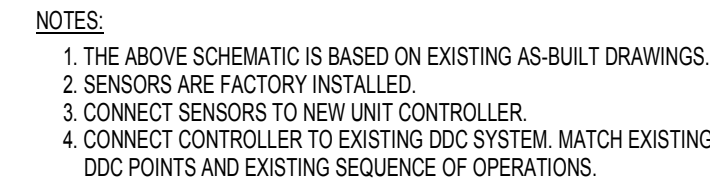
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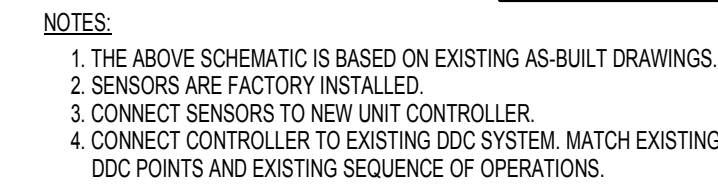
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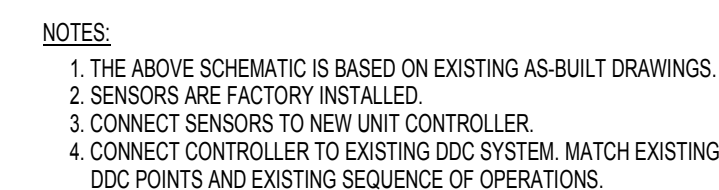
M-103



SCALE: NO SCALE



SCALE: NO SCALE



SCALE: NO SCALE

ISSUE LOG

PROFESSIONAL SEALS:

Thomas J. Farmer
FL PE # 58890

SHEET TITLE:

HVAC CONTROLS

SHEET INFORMATION:	
JOB No. 100040229	Date Issued: 2/10/15
Modeled By: REP	Sheet Number: M-201
Checked By: DLH	
QC Review: TJF	
Phase: CD	

M-301

Orange County Regional History Center

65 E Central Blvd, Orlando, FL 32801

100040229

TO THE BEST OF THE ENGINEER'S KNOWLEDGE, THE PLANS AND SPECIFICATIONS COMPLY WITH THE FLORIDA BUILDING CODE 2010.

No.	Date	Description

ISSUE LOG

PROFESSIONAL SEALS:

Wendy Bonnett
FL P.E. #62389

SHEET TITLE:

STRUCTURAL
SUPPORT DETAILS

SHEET INFORMATION:		
JOB No. 100040229	Date Issued: 2/10/15	
Modelled By: LBB	Sheet Number:	
Checked By: WJB		
QC Review: SA		
Phase: CD		

S-500

STRUCTURAL STEEL

- A. ALL HOT ROLLED STEEL PLATES, SHAPES, SHEET PILING, AND BARS SHALL BE NEW STEEL CONFORMING TO ASTM SPECIFICATION A6-98A.
- B. STRUCTURAL STEEL SHALL BE AS FOLLOWS, U.N.O.:
1. WIDE FLANGE SHAPES ASTM A992 FY = 50 KSI
 2. STRUCTURAL RECTANGULAR TUBING ASTM A500 GRADE B FY = 46 KSI
 3. ALL OTHER STRUCTURAL STEEL ASTM A36FY = 36 KSI
4. CONNECTION MATERIALS:
- a. BEAM COLUMN STIFFENER PLATES AND DOUBLER PLATES: ASTM A572 GRADE 50
 - b. ALL OTHER CONNECTION MATERIAL, U.N.O.: ASTM A36 UNLESS A HIGHER GRADE OF STEEL IS REQUIRED BY STRENGTH AND PROVIDED THE RESULTING SIZES ARE COMPATIBLE WITH THE CONNECTED MEMBERS.
8. ASTM A572 GRADE 50 IS ACCEPTABLE AS A SUBSTITUTE FOR A992
- C. STRUCTURAL STEEL SHALL MEET THE LATEST AISC "SPECIFICATION FOR THE DESIGN, FABRICATION AND ERECTION OF STRUCTURAL STEEL FOR BUILDINGS".
- D. THE CENTERLINES OF ALL COLUMNS AND BEAMS SHALL BE LOCATED ON COLUMN LINES UNLESS OTHERWISE SHOWN.
- E. CONNECTIONS:
1. BOLTS SHALL BE A325N TYPE 1, UNLESS OTHERWISE INDICATED.
 2. ALL BOLTS SHALL BE SNUG TIGHT U.N.O. BOLTS SHALL BE TIGHTENED UNTIL ALL PLIES OF THE JOINT ARE IN FIRM CONTACT.
 3. WELDING SHALL CONFORM TO THE STANDARDS SET FORTH IN AWS PUBLICATION, "WELDING IN BUILDING CONSTRUCTION".
 4. UNLESS NOTED OTHERWISE, ELECTRODES FOR WELDING SHALL CONFORM TO E70XX (SMAW), F7XX-EXXX (SAW), ER70S-X (GMAW), OR E7XT-X (FCAW). WEATHERING STEEL ELECTRODES SHALL CONFORM TO THE ANSI/AWS D1.1 MANUAL. ELECTRODES FOR GRADE 60 OR GRADE 65 MATERIAL SHALL CONFORM TO E80XX (SMAW), F8XX-EXX-XX (SAW), ER80S-X (GMAW), OR E8XT-X (FCAW).
 5. WHERE THE WORK OF OTHER TRADES REQUIRES CUTS, HOLES, ETC., IN STRUCTURAL STEEL MEMBERS, CUTS, HOLES, ETC., SHALL BE MADE IN THE SHOP AND SHALL BE SHOWN ON THE SHOP DRAWINGS. MAKING HOLES OR CUTS IN STRUCTURAL STEEL MEMBERS IN THE FIELD WILL NOT BE PERMITTED WITHOUT SPECIFIC APPROVAL OF THE ENGINEER.
- F. ALL STRUCTURAL STEEL SHALL BE SHIPPED WITH ONE COAT OF SHOP PRIMER EXCEPT THOSE MEMBERS THAT ARE GALVANIZED OR IN AREAS SCHEDULED TO RECEIVE FIRE PROOFING.



REMOVE BEAM
EACH SIDE

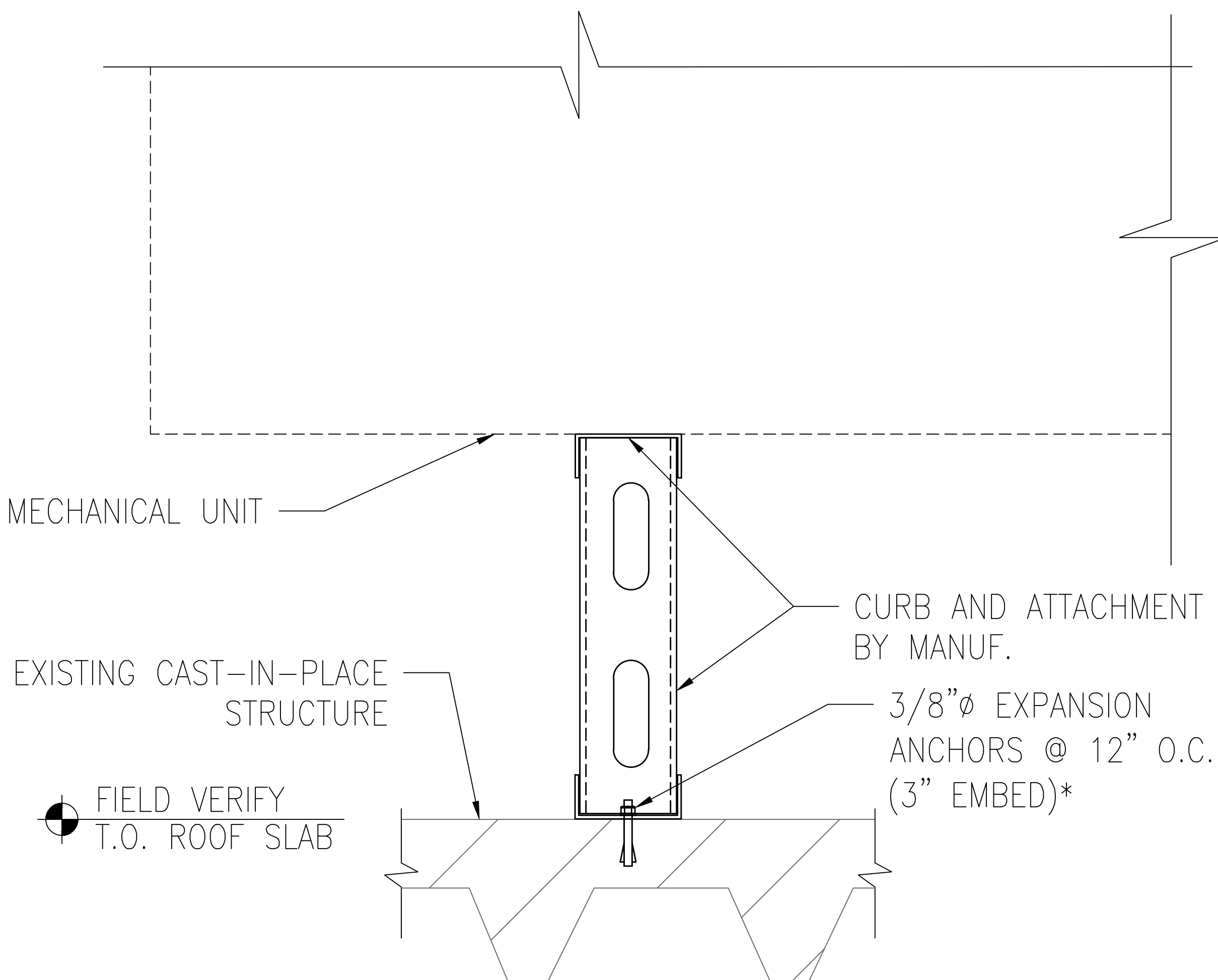
GRIND WELDS
OFF AND REMOVE
CHANNELS AND
PLATES ABOVE,
TYP.

PROTECT TOP FLANGE
OF EXISTING BEAM
FOR REUSE
TOUCH UP ANY
DAMAGED GALVANIZING

3 EXISTING COOLING TOWER SUPPORT

S-500

SCALE: N.T.S.



NOTE:

1. * FIELD-VERIFY SLAB THICKNESS. IF LESS THAN 4" THICK NOTIFY ENGINEER FOR REVISED ANCHOR DESIGN.

1 NEW MECHANICAL UNIT SUPPORT

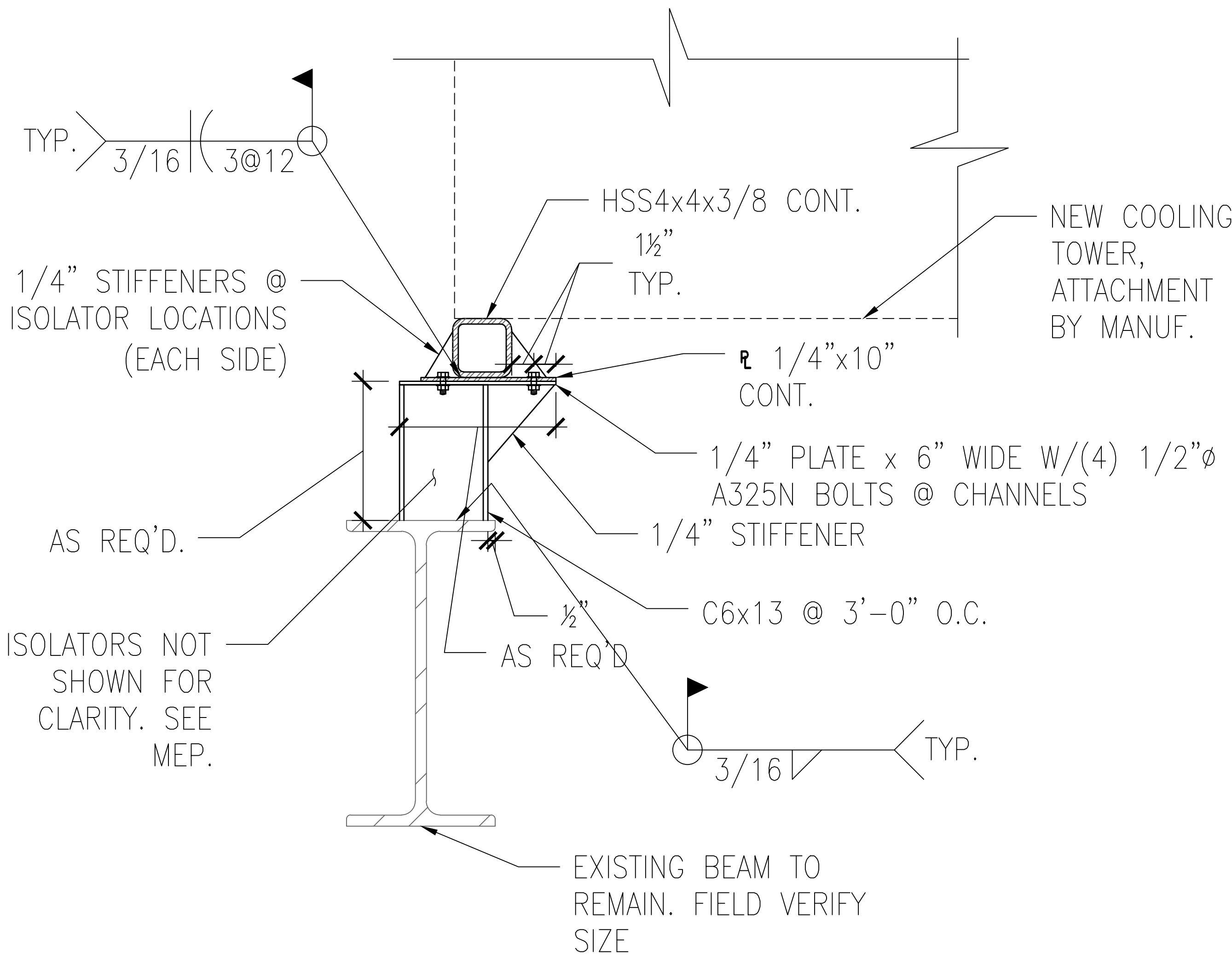
S-500

SCALE: N.T.S.

4 GENERAL NOTES

S-500

SCALE: N.T.S.



NOTES:

1. GALVANIZE ALL STEEL.
2. DEMOLISH EXISTING BEAM & BEAM SUPPORT CHANNELS.

2 COOLING TOWER SUPPORT

S-500

SCALE: N.T.S.